Optimizing Context Retrieval

A Comparative Study of RAG vs. Zero-Shot LLM in Digital Art Galleries

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Motivation, Main Idea, & Results

- **Motivation:** Large Language Models (LLMs) frequently suffer from **hallucinations** when queried about domain-specific, private, or niche datasets such as a personal digital art portfolio.

- **Main Idea:** Implement a Retrieval-Augmented Generation (RAG) architecture. By storing artwork metadata in a vector database, the system retrieves accurate background knowledge before generating a response, rather than relying solely on the LLM's pre-trained memory.

- **Results:** The experiment demonstrates that the RAG pipeline significantly increases the accuracy of domain-specific queries and effectively eliminates AI hallucinations compared to a standard Zero-Shot LLM approach.

Literature Review

- Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

  https://arxiv.org/abs/2005.11401

  - RAG successfully combines parametric memory with non-parametric memory, allowing AI models to dynamically access domain-specific or updated knowledge without retraining.

- Survey of Hallucination in Natural Language Generation

  https://arxiv.org/abs/2202.03629

  - LLMs naturally hallucinate when dealing with closed-domain data. Context injection—retrieving and providing factual documents within the prompt—is proven to be the most cost-effective and reliable method to mitigate these errors.

Approach Used

- **Dataset:** A custom JSON/Dictionary dataset representing a digital art gallery (including artwork titles, mediums, and conceptual descriptions).
- **Vector Database:** ChromaDB (a lightweight, local open-source vector database) is used to generate embeddings from the artwork descriptions and perform similarity matching based on the user's query.
- **Generative AI:** Google Gemini API (gemini-1.5-flash) is used for the final natural language synthesis.
- **Experiment Setup (The Comparison):**
  - Baseline (Zero-Shot): Asking Gemini directly about a specific digital artwork.
  - RAG System:
    - Querying ChromaDB for the most relevant artwork description.
    - Injecting the retrieved text into the prompt.
    - Generating the answer via Gemini.

System Demonstration

- Video Highlights:
  - Demonstrates the Zero-Shot model fabricating incorrect information about an artwork.
  - Showcases the RAG system accurately retrieving the hidden context and generating a precise, factual explanation.
- Watch the Full Demonstration:

  https://youtu.be/CVQQC29FnYo

Code & Resources

- Source Code: https://github.com/ShaoEH/ai-hw-spring-2026-final-project
- Technologies & Relevant Links:
  - Python 3
  - ChromaDB (Vector Database)
  - Google Generative AI SDK

Thank You